C PROGRAMMING WITH LINUX (UBUNTU)

Bui Ngoc Bao Tran – QE170086

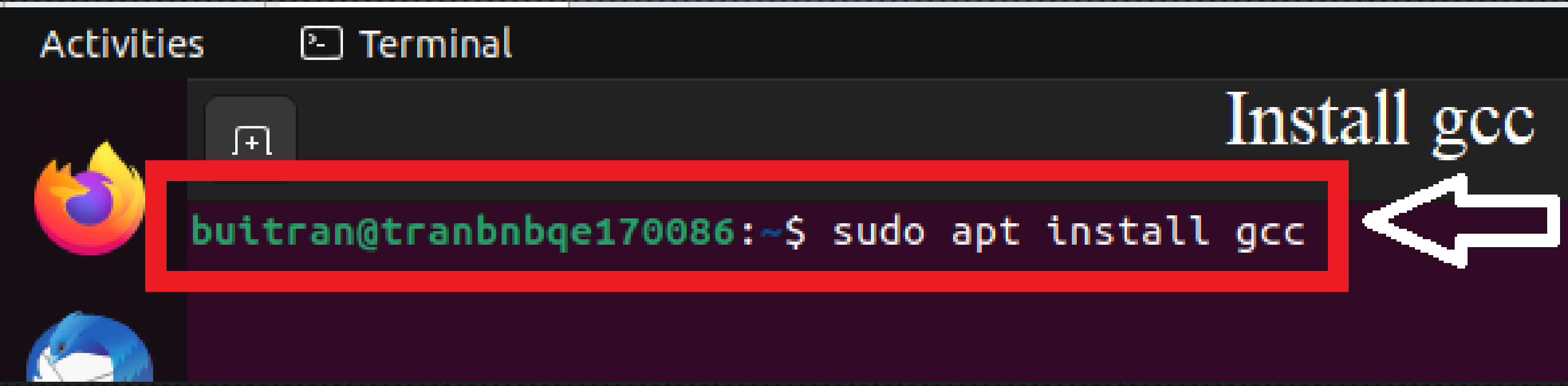
Before starting, you have to download VMWare and Ubuntu Virtual machine.

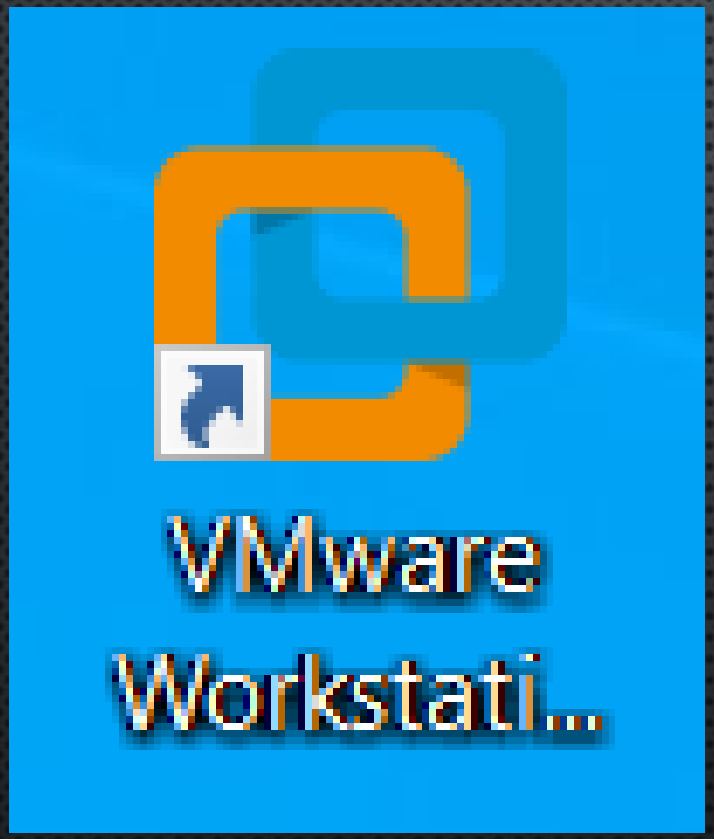
If you haven't download it, you can install VMWare at: https://www.vmware.com/asean/products/workstation-pro/workstation-pro-evaluation.html

And download Ubuntu at: https://ubuntu.com/download/desktop

Make sure you install "gcc" to execute the code in our exercise.

Install “gcc” to easily excute C programme in Linux (Ubuntu).





Access to the VMWare icon on your desktop.

Click "***Power on this virtual machine***" to start Ubuntu.

Home | Ubuntu

Ubuntu

Power on this virtual machine

Edit virtual machine settings

Click here to open Virtual machine.



Devices

Memory	4 GB
Processors	2
Hard Disk (SCSI)	20 GB
CD/DVD 2 (SATA)	Using file E:\PHA...
CD/DVD (SATA)	Using file autoin...
Floppy	Using file autoin...
Network Adapter	NAT
USB Controller	Present
Sound Card	Auto detect
Printer	Present
Display	Auto detect

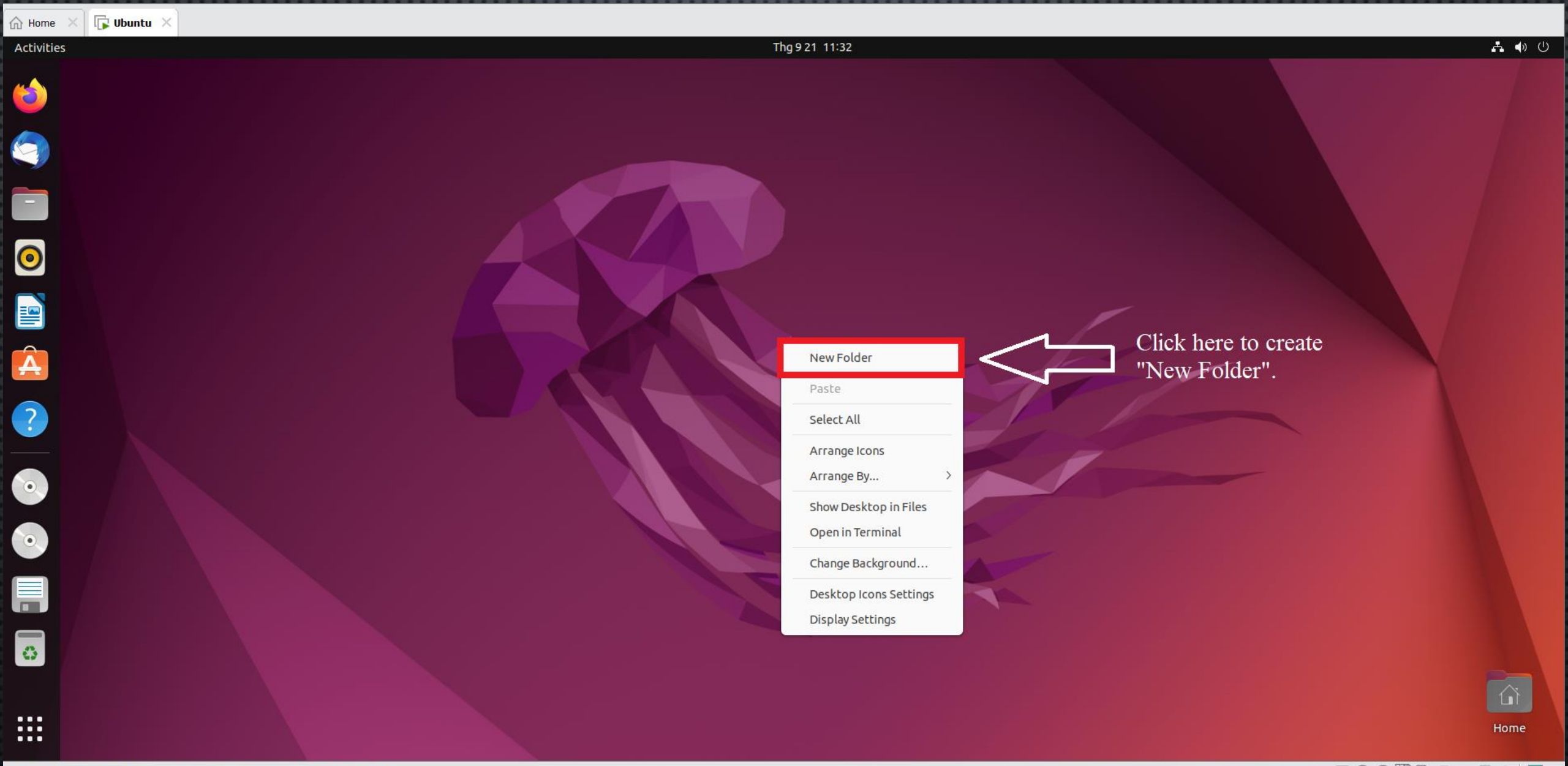
Description

Type here to enter a description of this virtual machine.

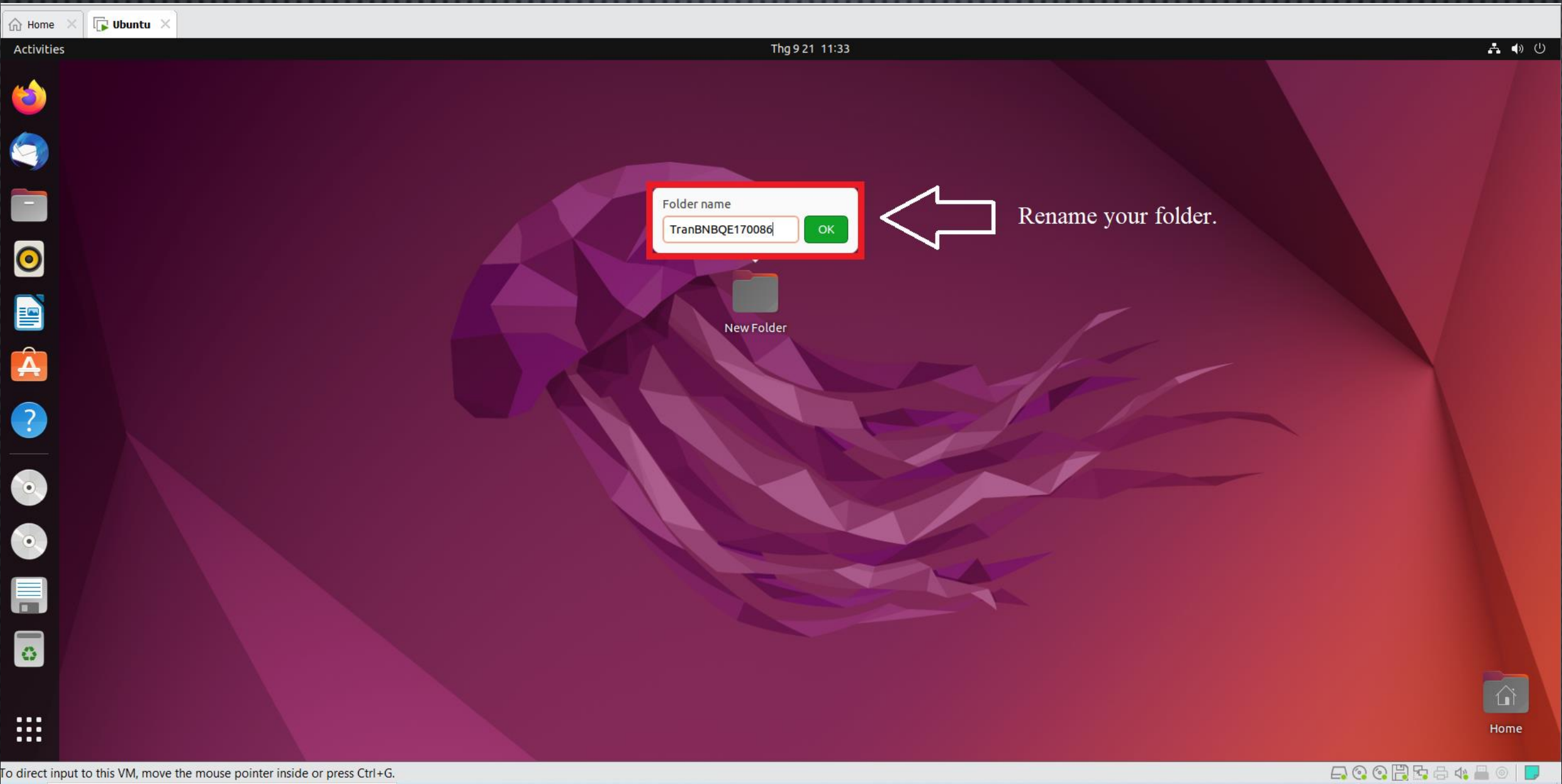
Virtual Machine Details

State: Powered off
Configuration file: E:\PHAN MEM\UBUNTU\Ubuntu.vmx
Hardware compatibility: Workstation 16.2.x virtual machine
Primary IP address: Network information is not available

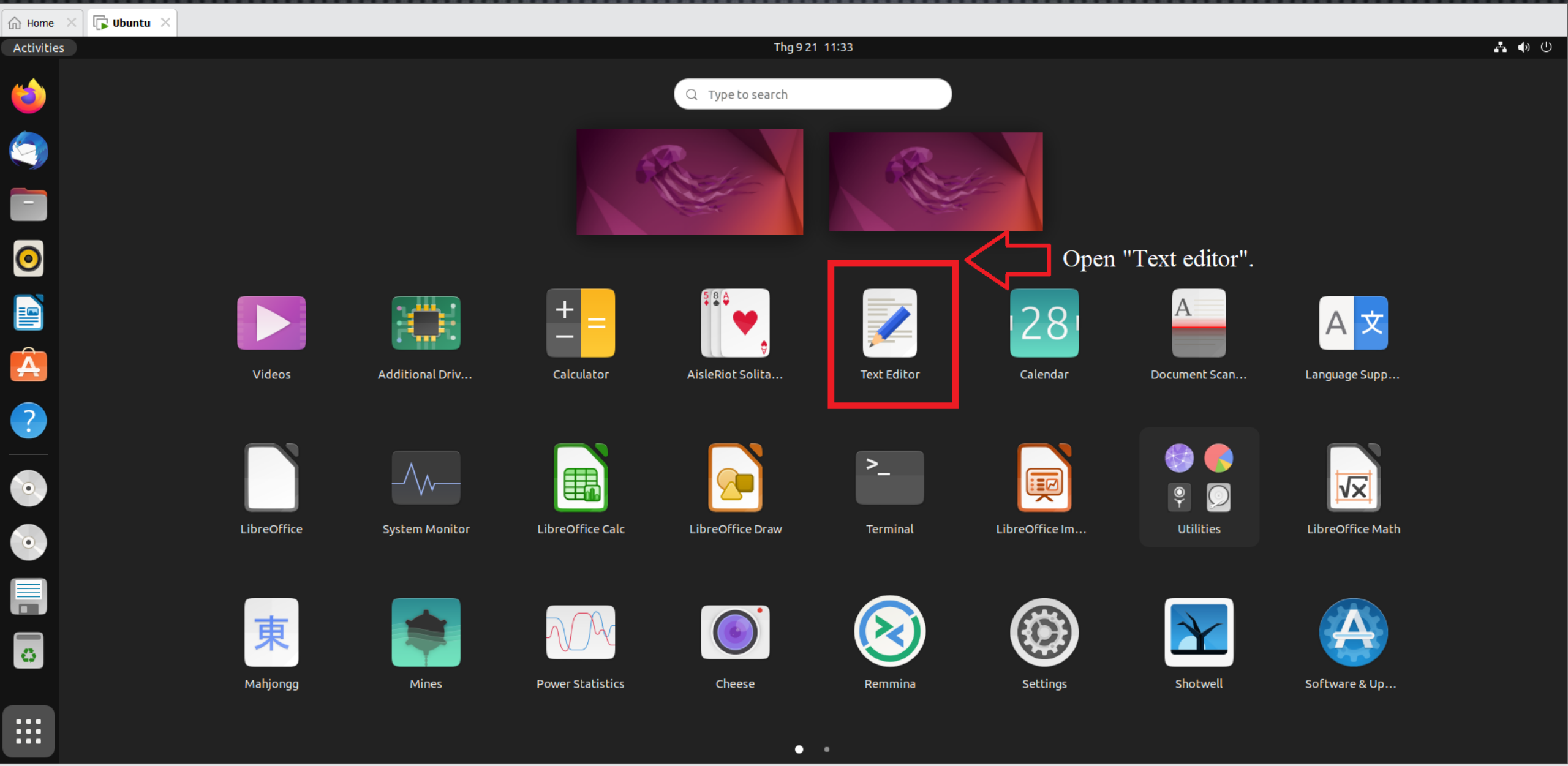
Rightclick mouse and choose "***New Folder***" to create folder.



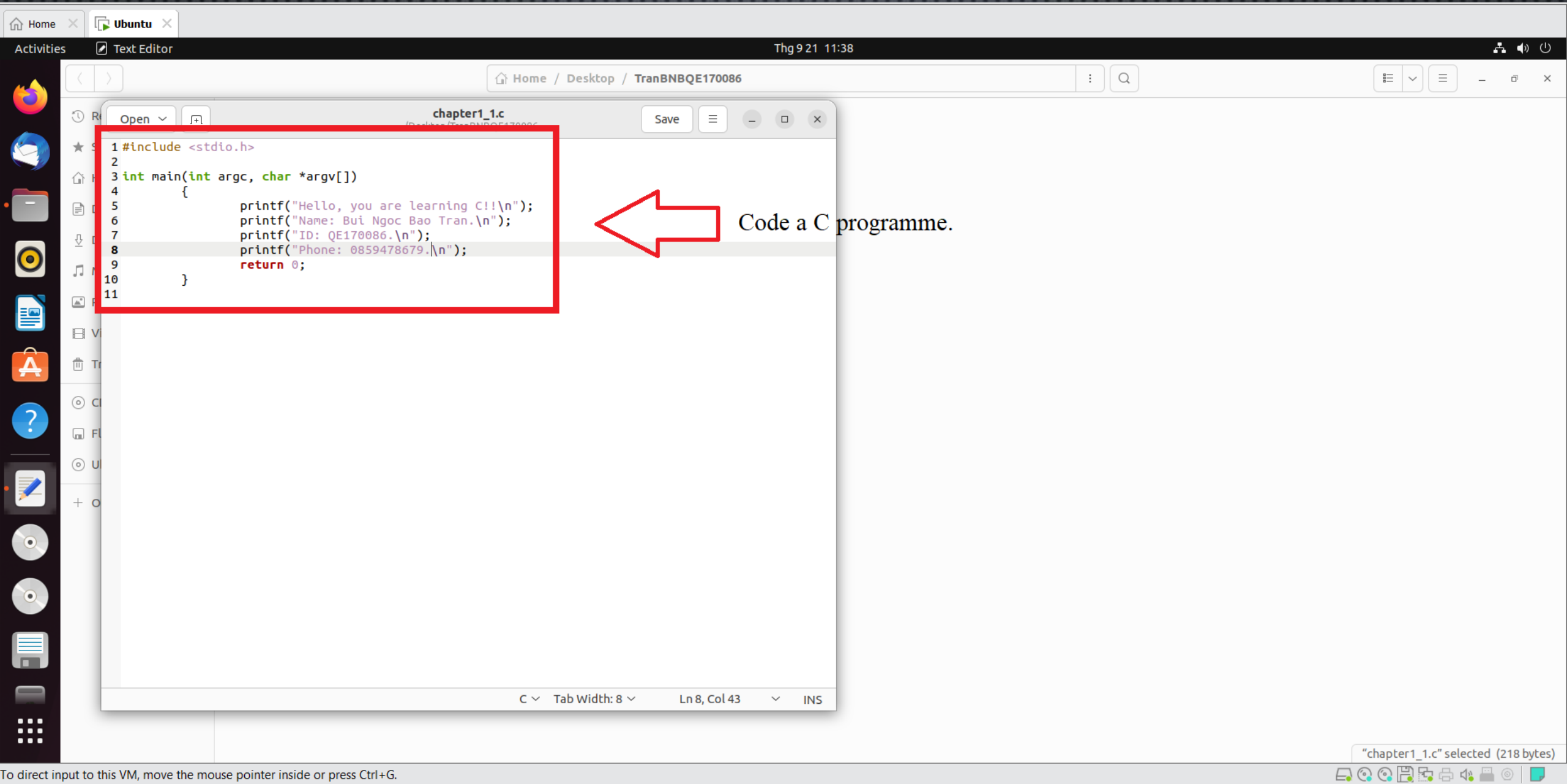
Rename your folder and choose "***OK***".



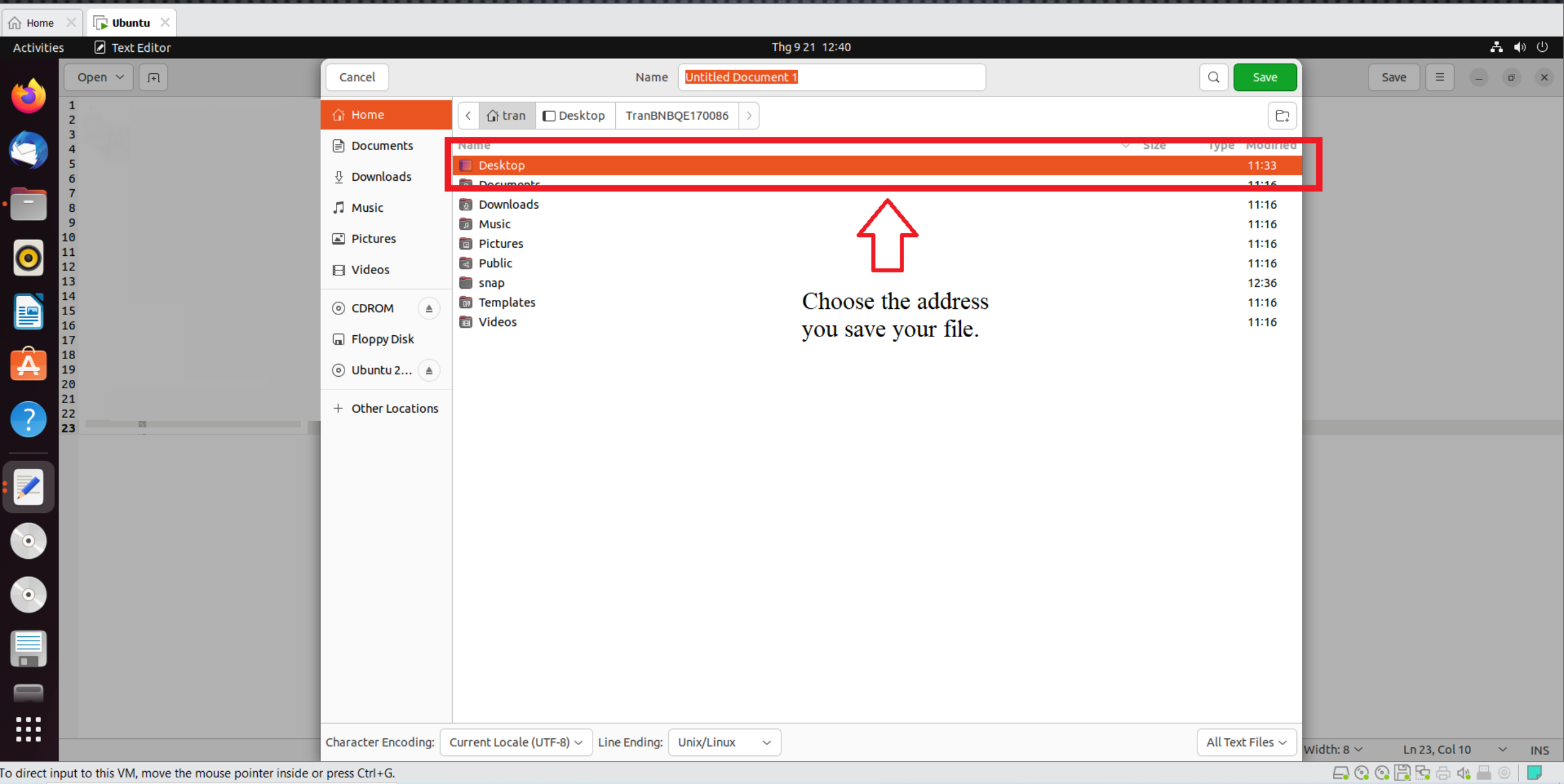
Open "***Text editor***".



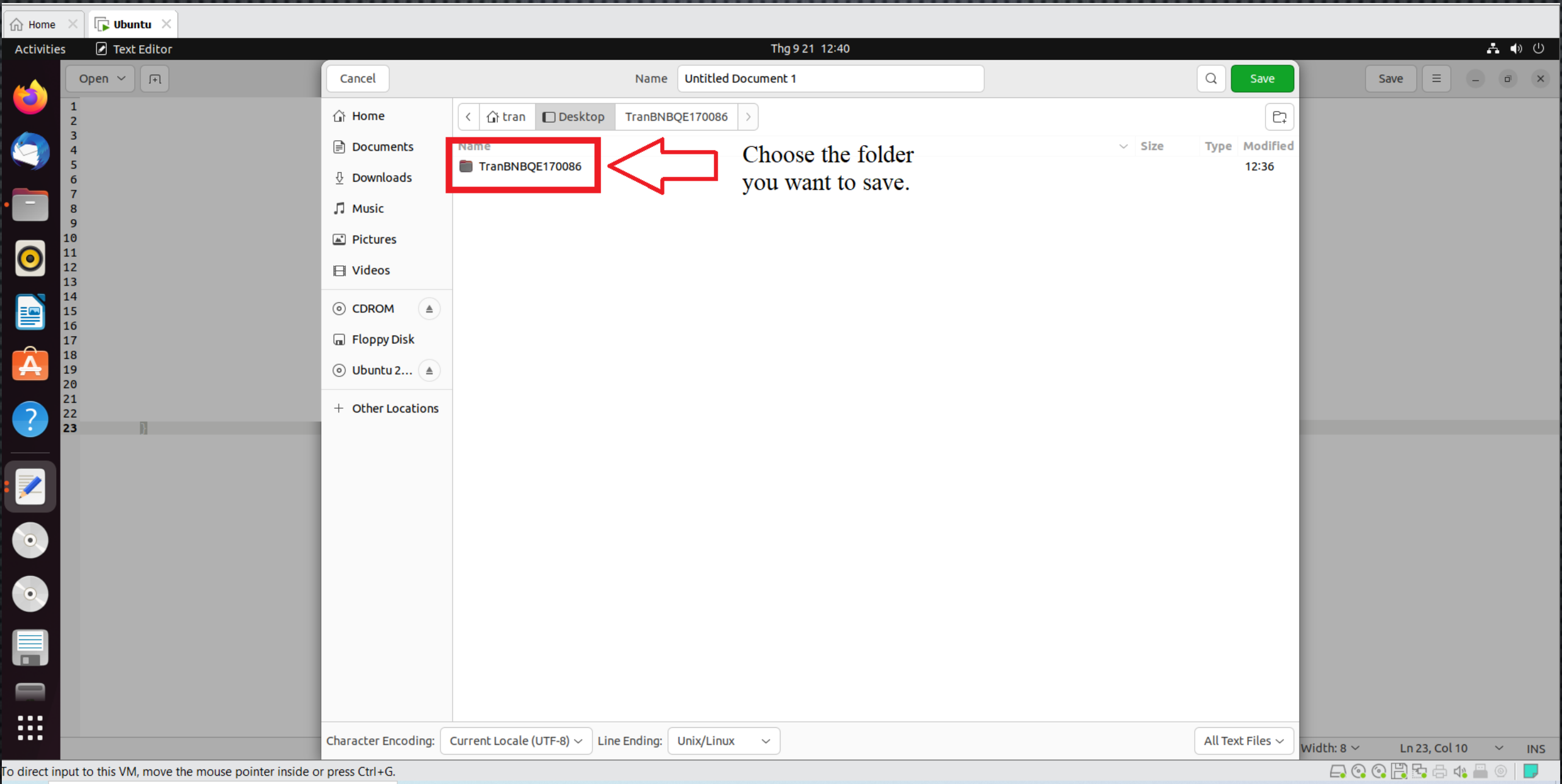
Code exercise 1 C programme.



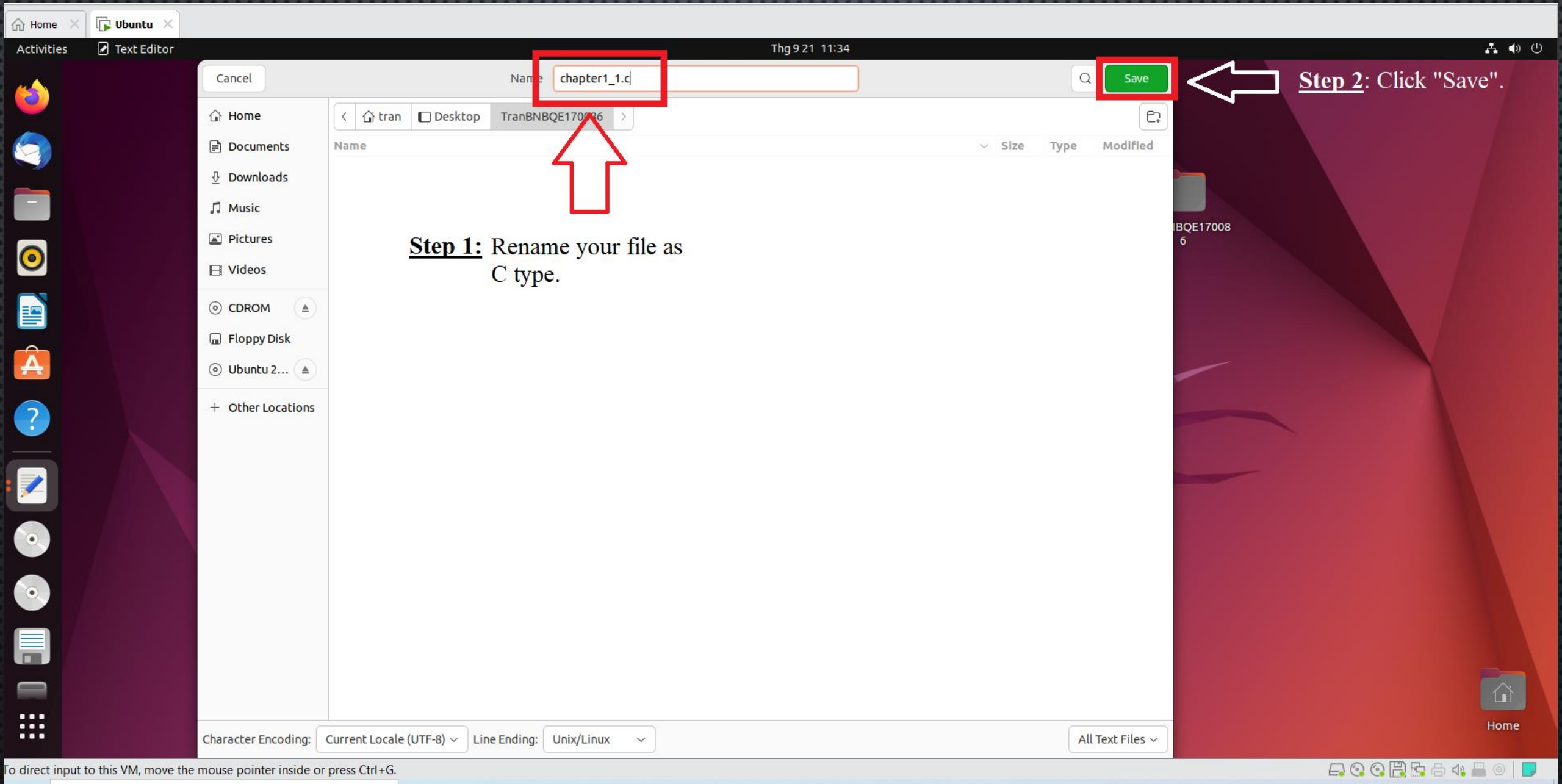
Choose the address.



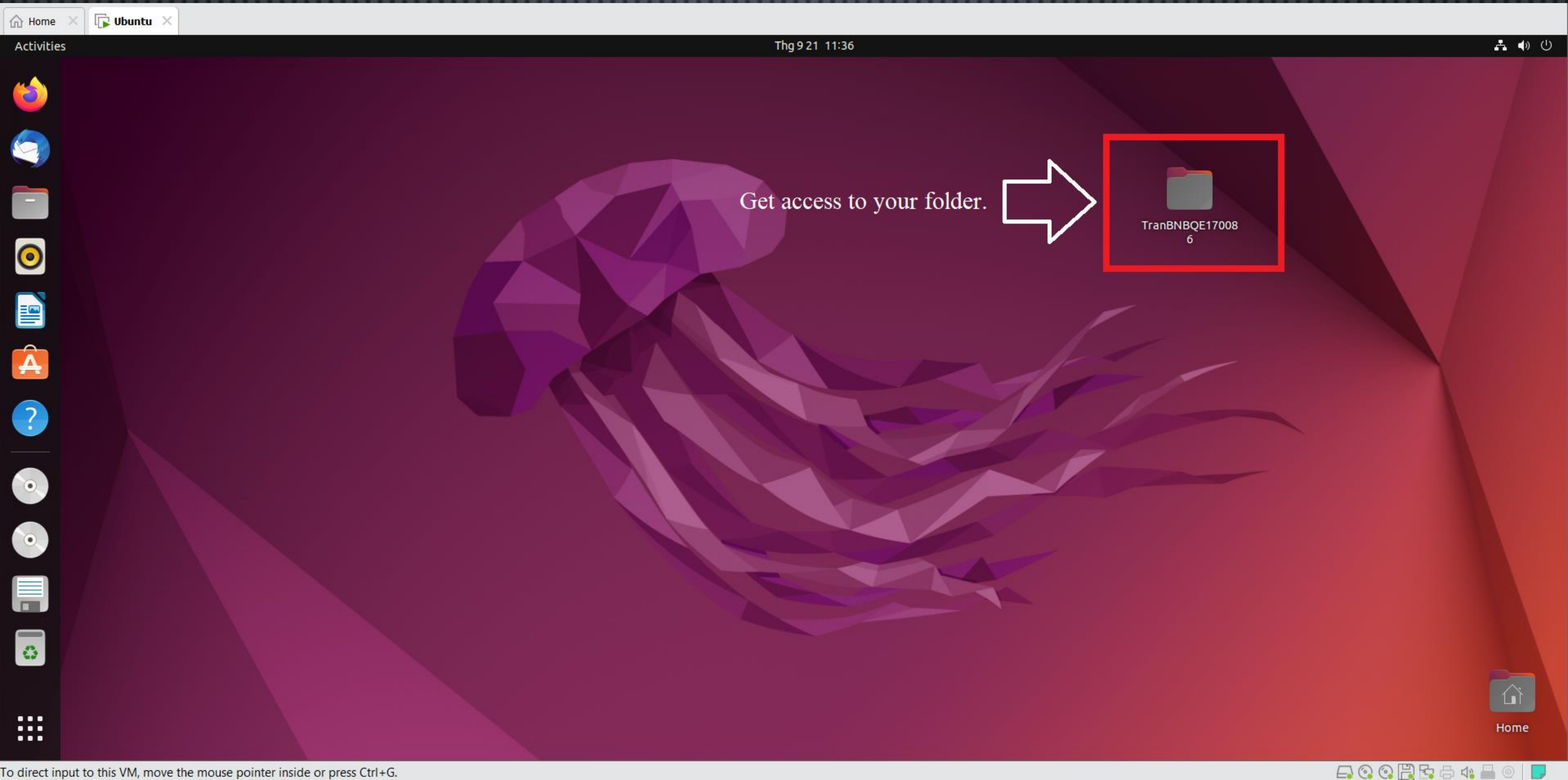
Select the destination.



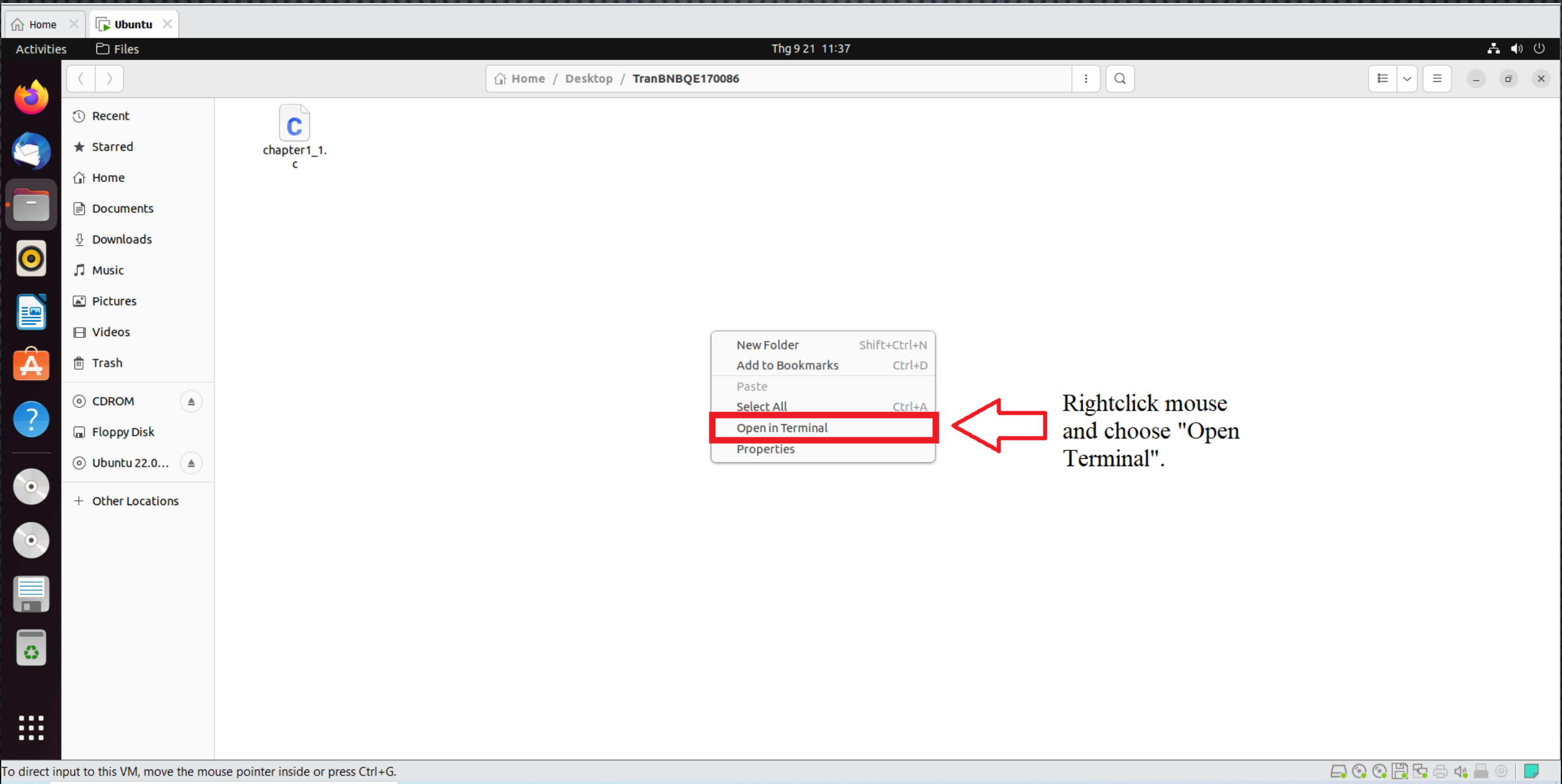
Rename your file and choose "***Save***".



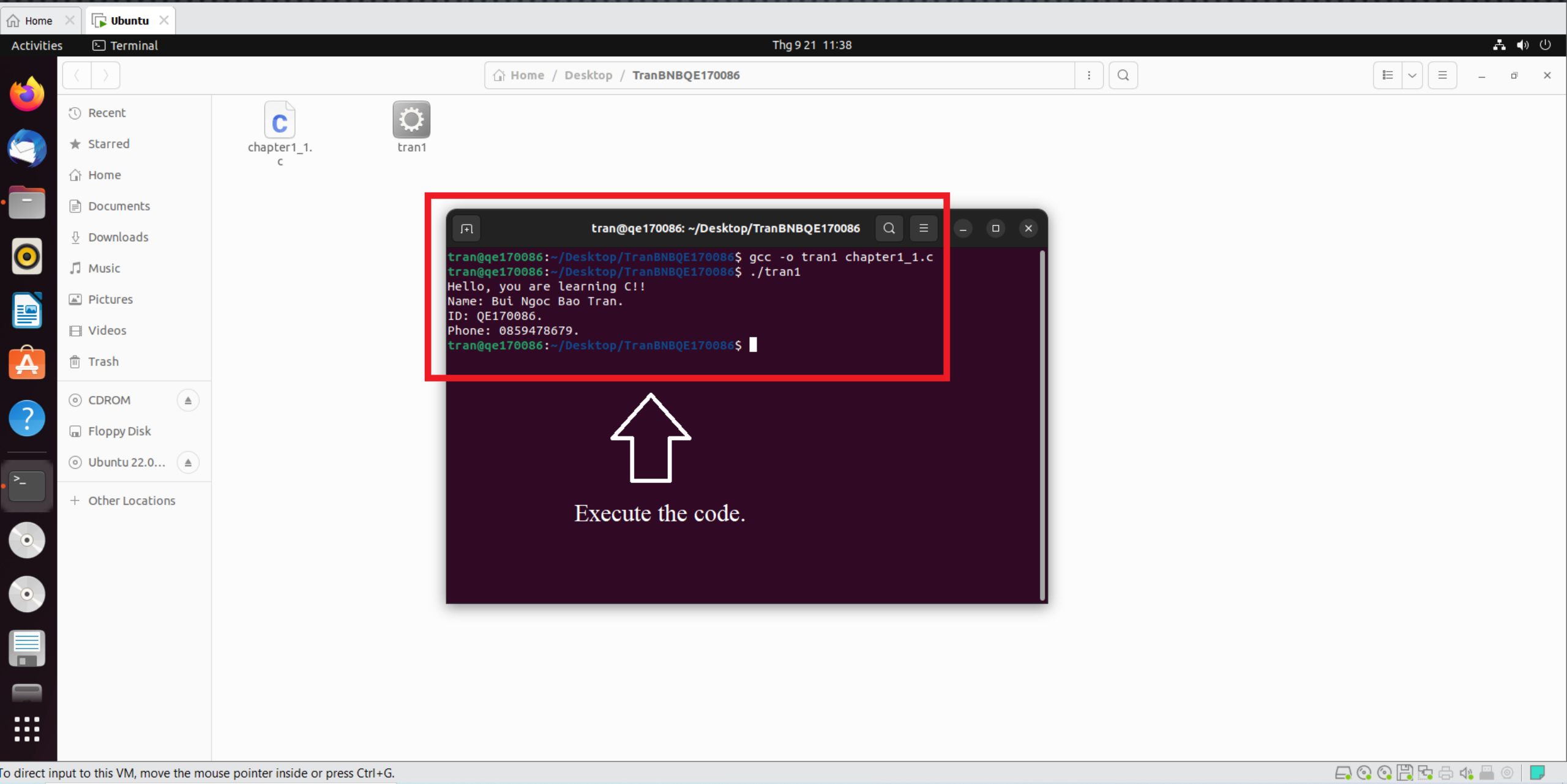
Doubleclick to access your new file.



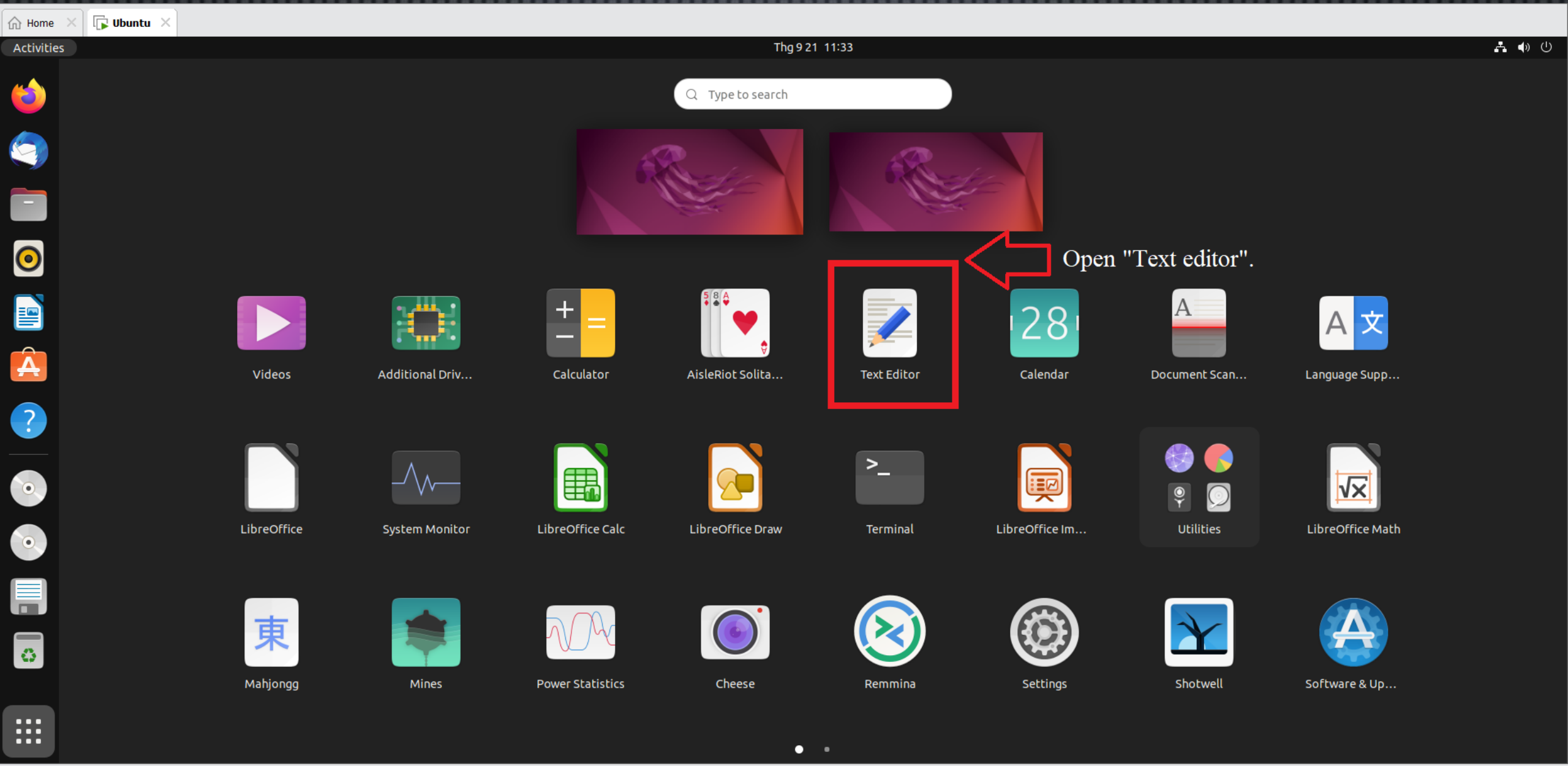
Rightclick mouse and choose "***Open in Terminal***".



Execute the code.



Open "***Text editor***".



Code exercise 2 C programme.

Home | Ubuntu

Activities | Text Editor | Thg 9 21 11:41

Open | *Untitled Document 1 | Save

```c
#include <stdio.h>

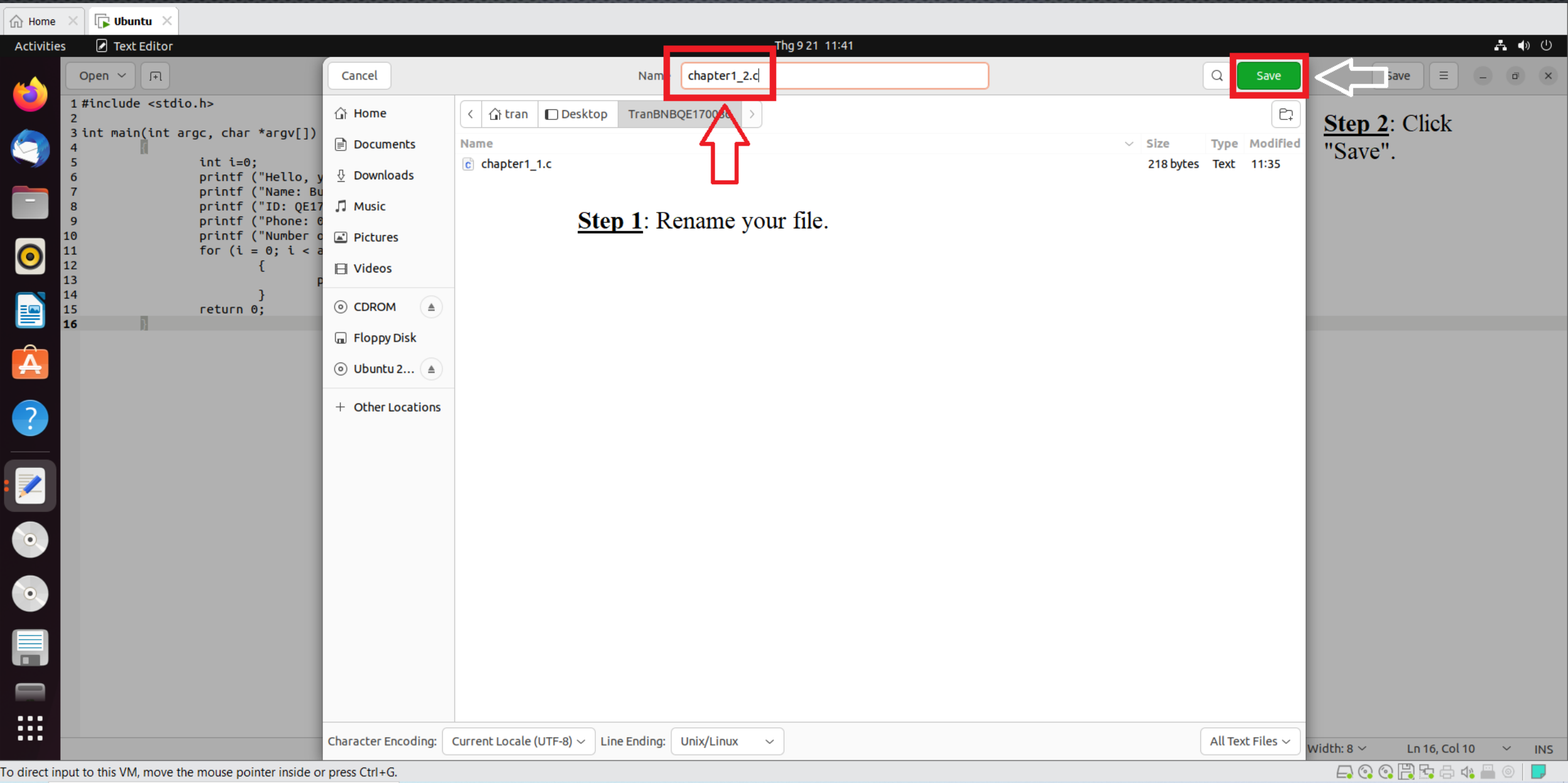
int main(int argc, char *argv[])
{
        int i=0;
        printf ("Hello, you are learning C !\n");
        printf ("Name: Bui Ngoc Bao Tran.\n");
        printf ("ID: QE170086.\n");
        printf ("Phone: 0859478679.\n");
        printf ("Number of arguments to the main function: %d\n", argc);
        for (i = 0; i < argc; i++)
                {
                        printf ("Argument number %d is %s\n", i, argv[i]);
                }
        return 0;
}
```

Code a C programme.

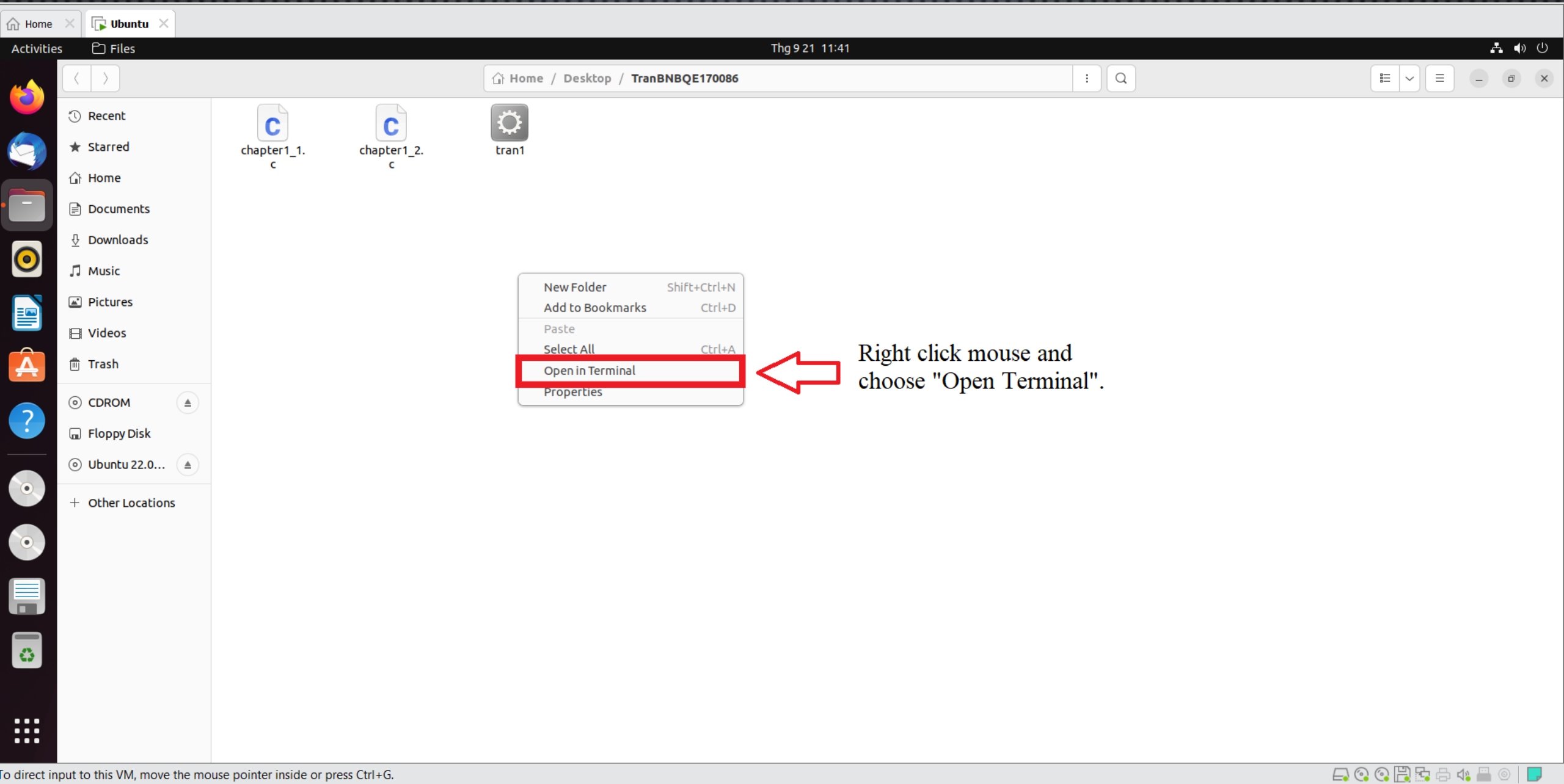
Plain Text | Tab Width: 8 | Ln 16, Col 10 | INS

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

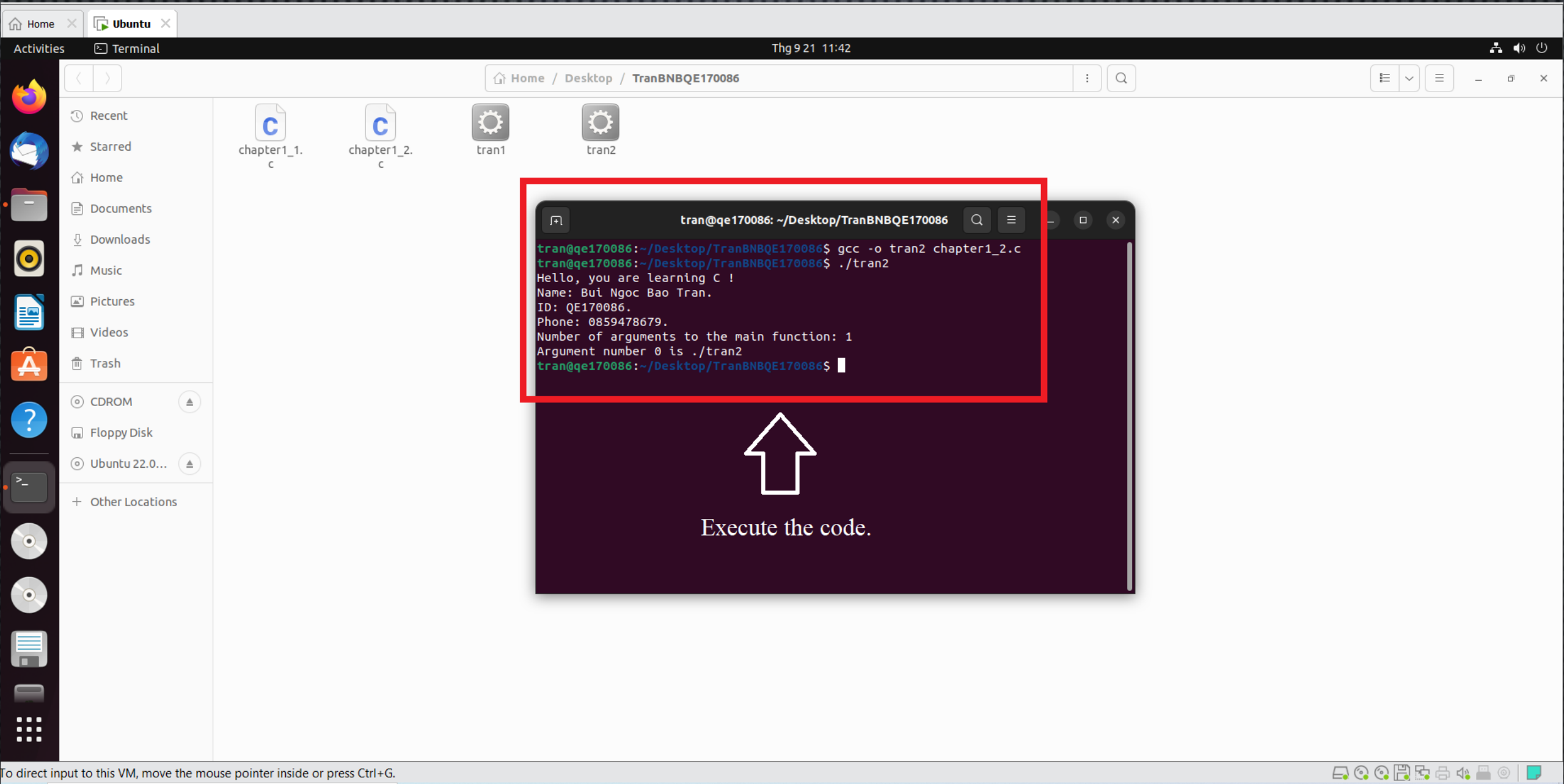
Rename your file and choose "***Save***".



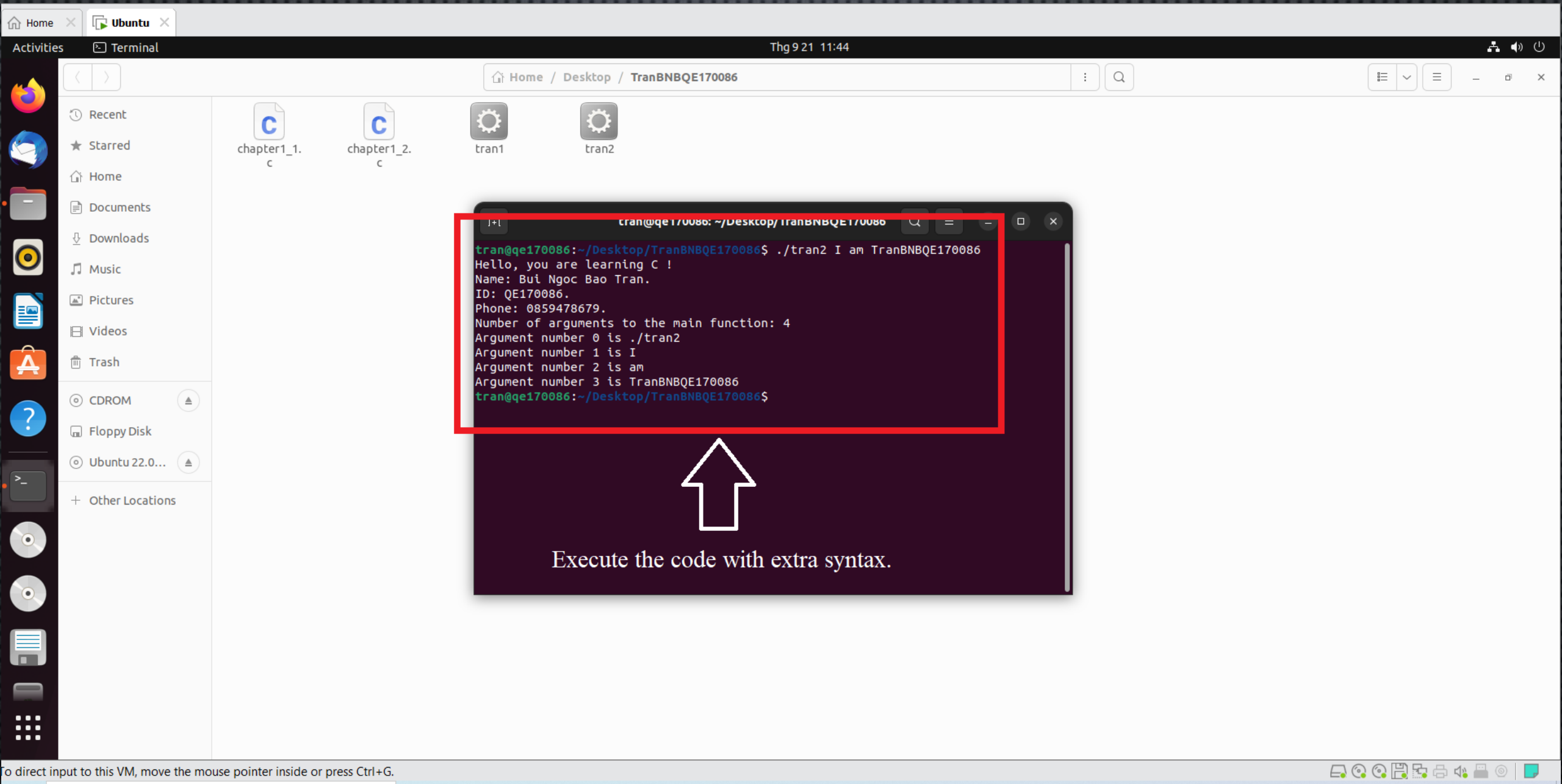
Rightclick mouse and choose "***Open in Terminal***".



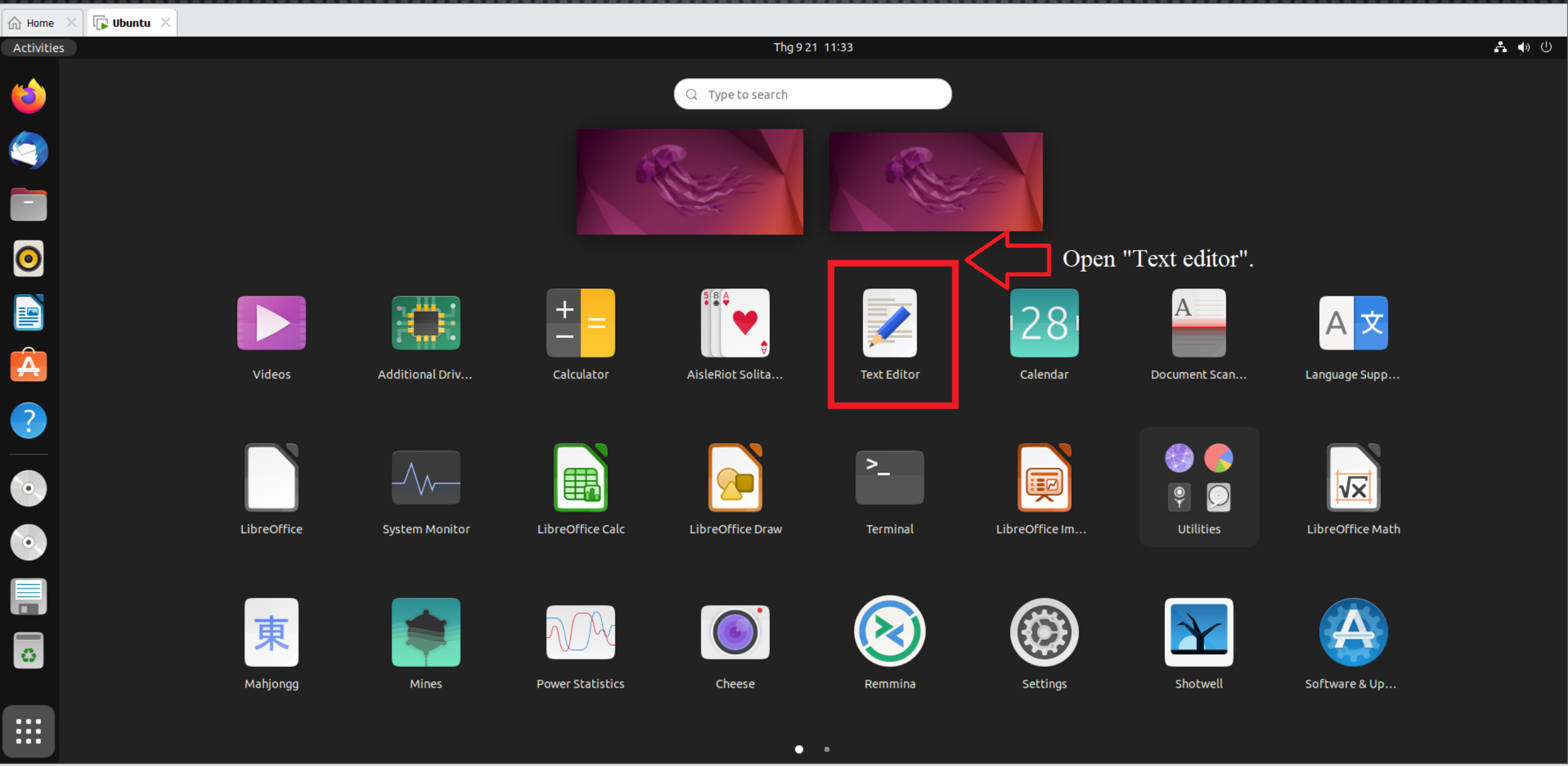
Execute the code.



Execute the code with an extra syntax.



Open "***Text editor***".



Code exercise 3 C programme.

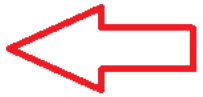
Home | Ubuntu

Activities | Text Editor | Thg 9 21 12:35

Open | *Untitled Document 1 | Save

```c
#include <stdio.h>

int main(int argc, char *argv[])
{
        int i = 0;
        printf ("Content-type: text/plain\n\n");
        printf ("Hello, you are still learning C !\n");
        printf ("Number of arguments to the main function: %d\n", argc);
        printf ("Name: Bui Ngoc Bao Tran.\n");
        printf ("ID: QE170086.\n");
        printf ("Phone: 0859478679.\n");
        for(i = 0; i < argc; i++)
                {
                        printf ("Argument number %d is %s\n", i, argv[i]);
                }
        return 0;
}
```

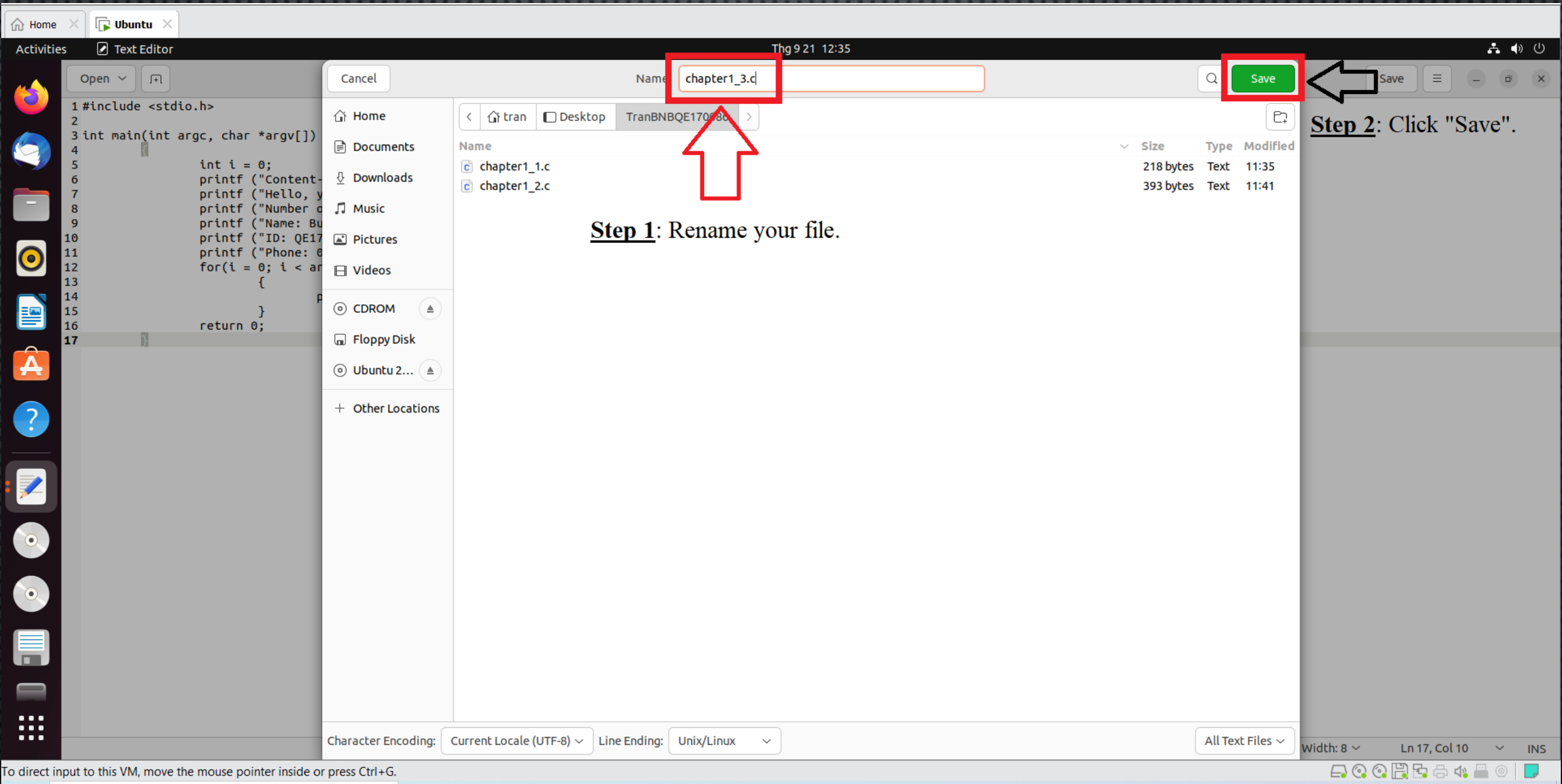


Code a C programme.

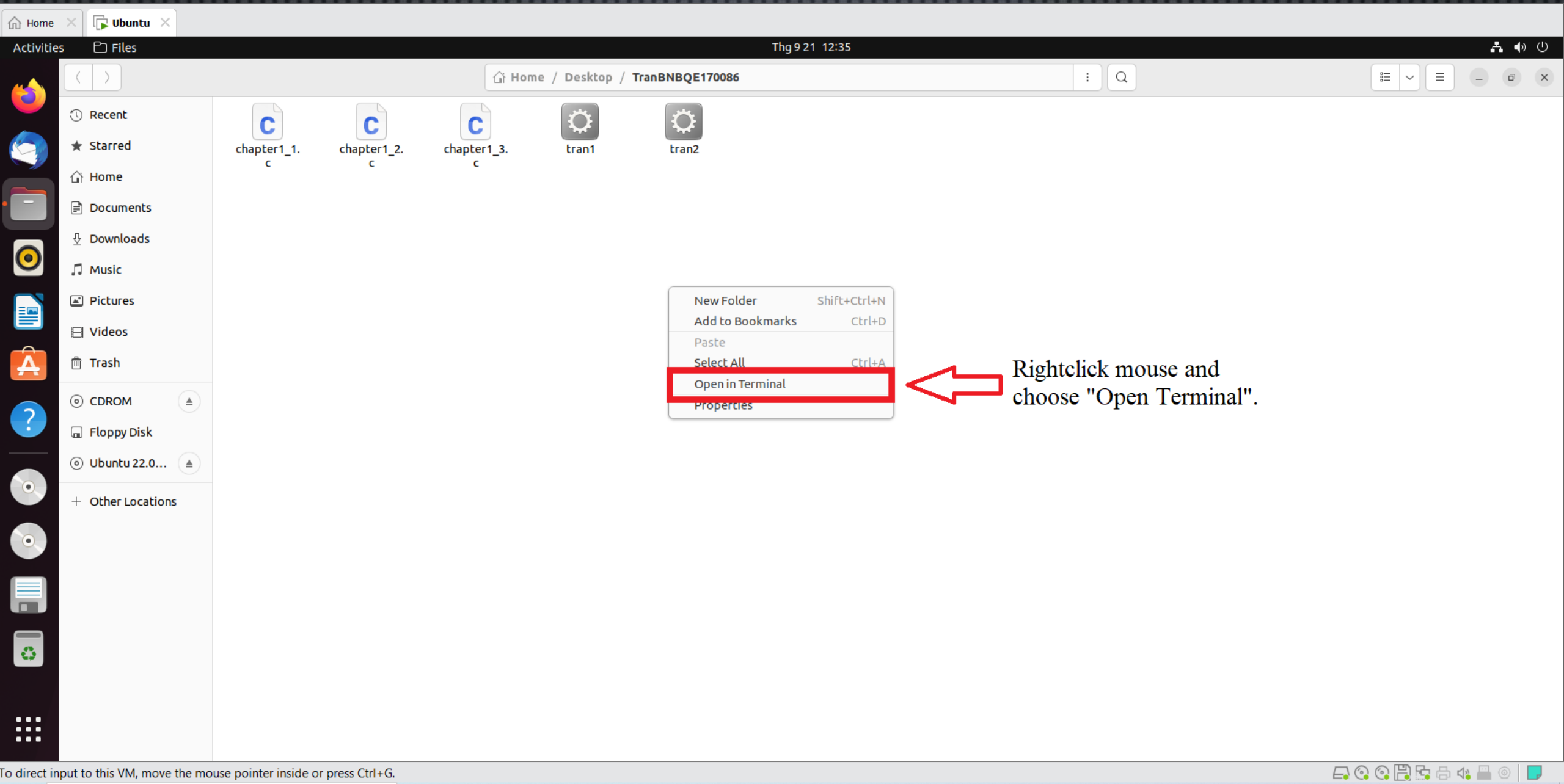
Plain Text | Tab Width: 8 | Ln 17, Col 10 | INS

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

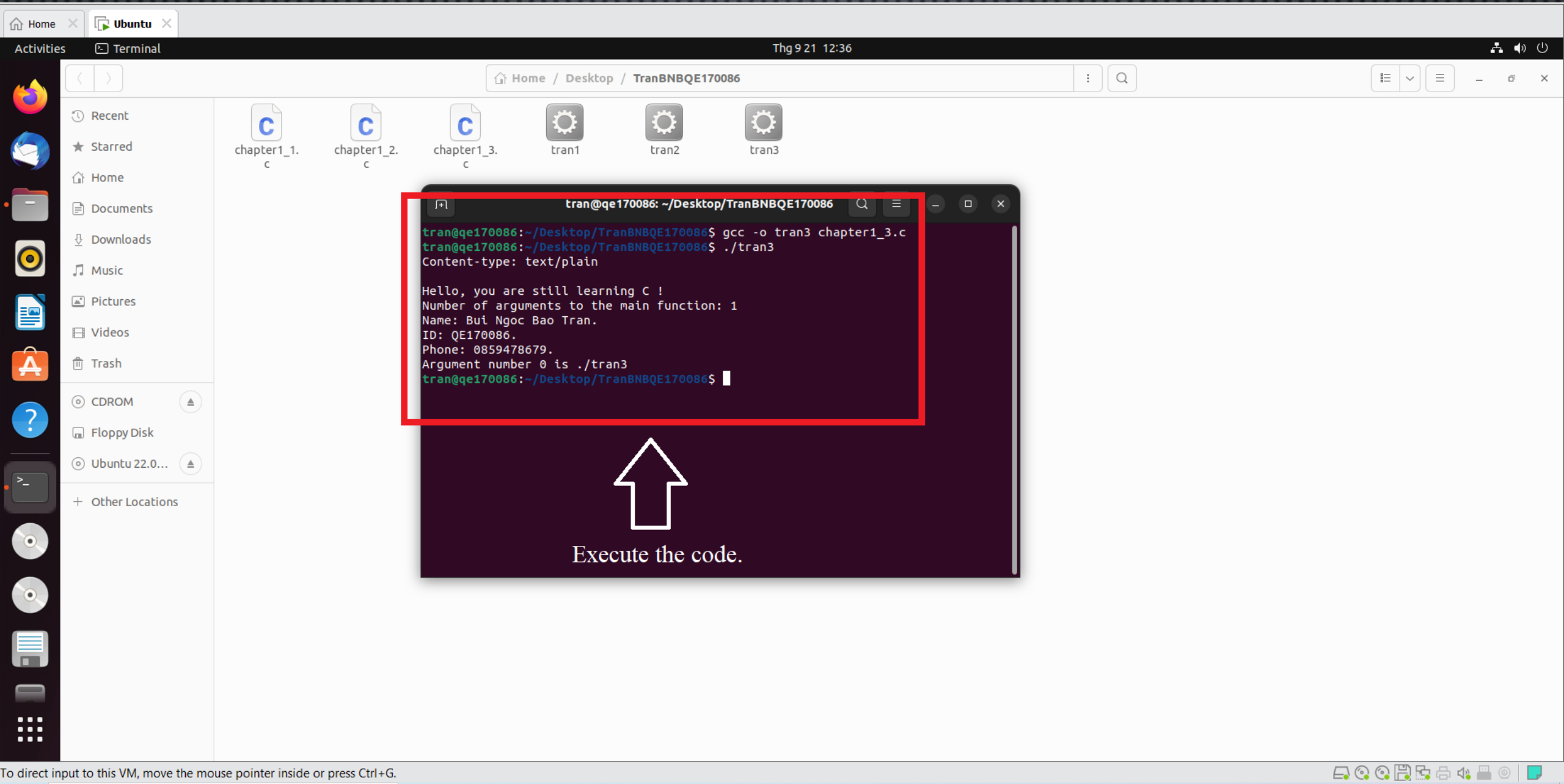
Rename your file and choose "***Save***".



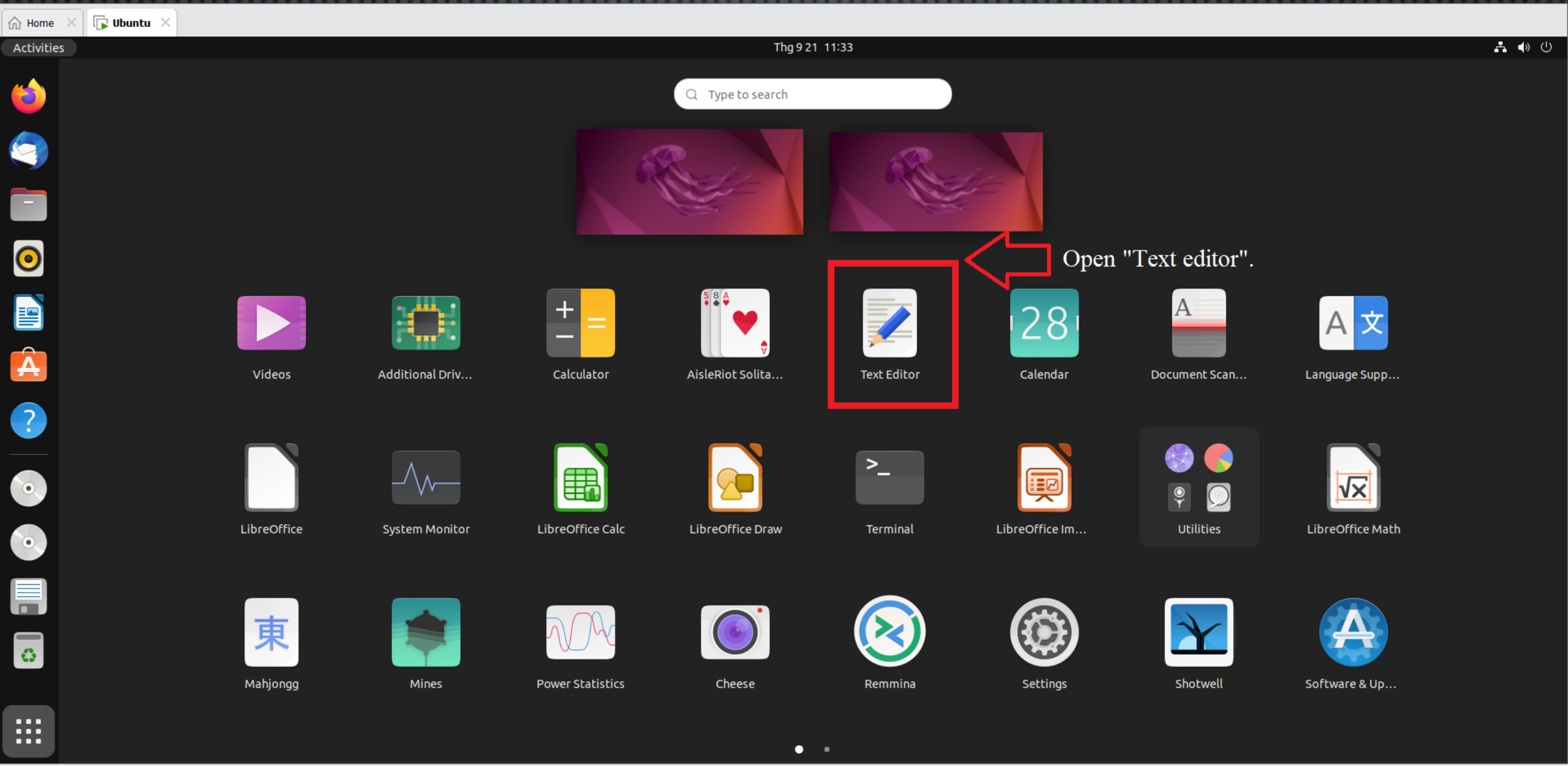
Rightclick mouse and choose "***Open in Terminal***".



Execute the code.



Open "***Text editor***".



Code exercise 4 C programme.

Home | Ubuntu

Activities | Text Editor | Thg 9 21 12:40

Open | *Untitled Document 1 | Save

```c
#include <stdio.h>

int main(int argc, char *argv[], char *env[])
{
        int i = 0;
        printf ("Content-type: text/plain\n\n");
        printf ("Hello, you are still learning C !\n");
        printf ("Number of arguments to the main function: %d\n", argc);
        printf ("Name: Bui Ngoc Bao Tran.\n");
        printf ("ID: QE170086.\n");
        printf ("Phone: 059478679\n");
        for(i = 0; i < argc; i++)
                {
                        printf ("Argument number %d is %s\n", i, argv[i]);
                }
        i = 0;
        printf ("Environment variables: \n");
        while(env[i])
                {
                        printf ("env[%d] = %s\n", i, env[i]); i++;
                }
        return 0;
}
```

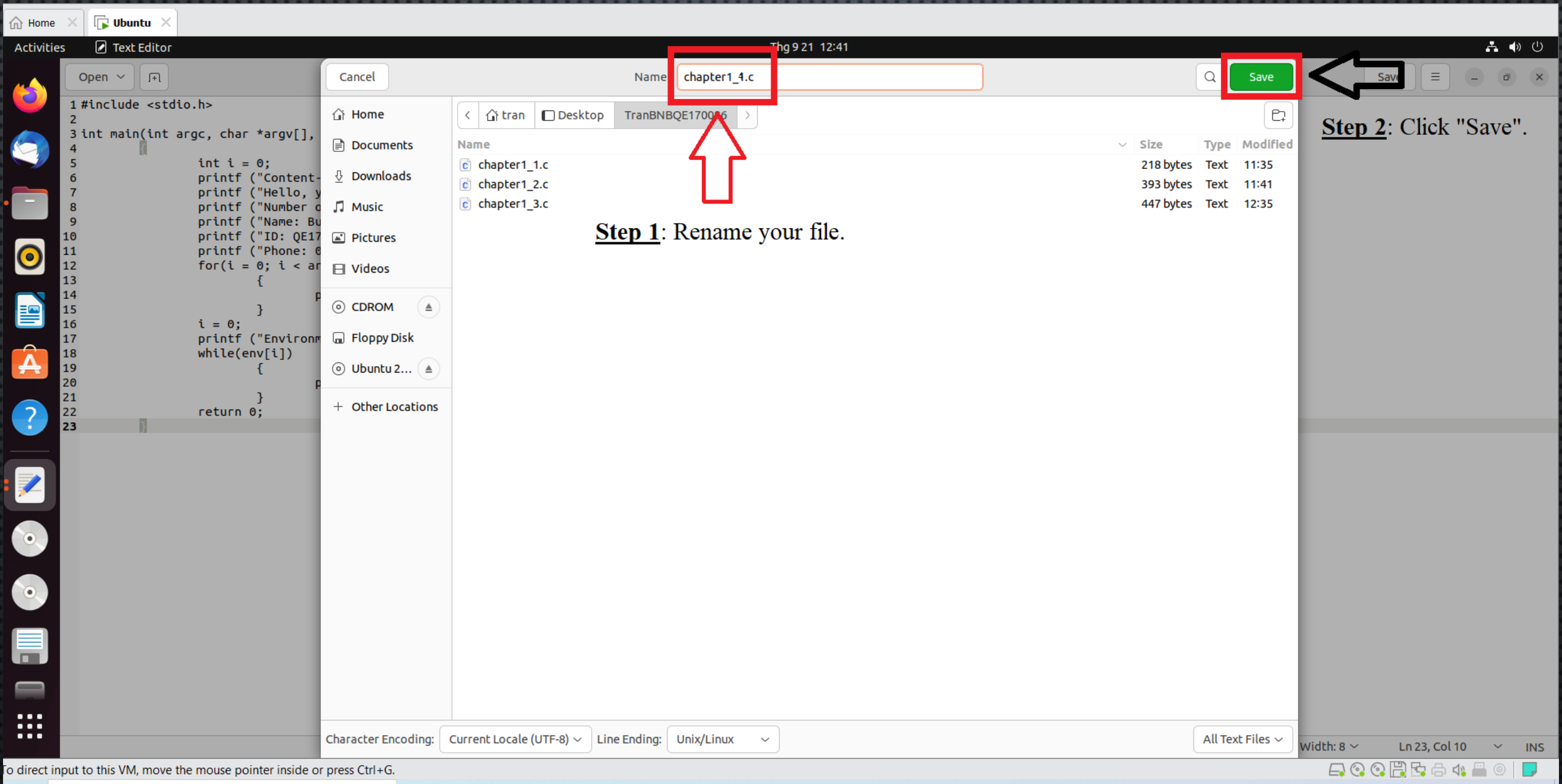
Code a C programme.

Bracket match found on line: 4

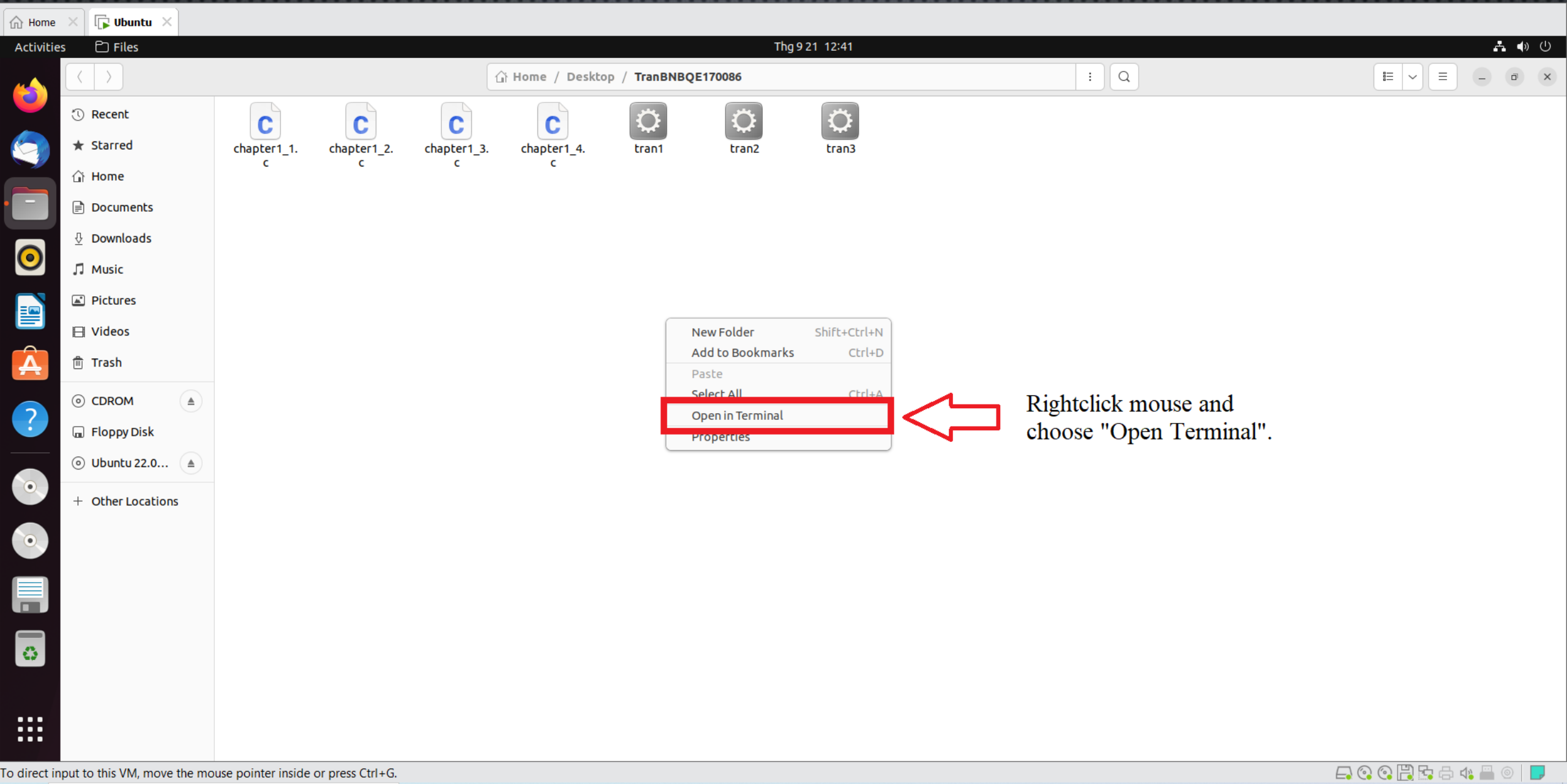
Plain Text | Tab Width: 8 | Ln 23, Col 10 | INS

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

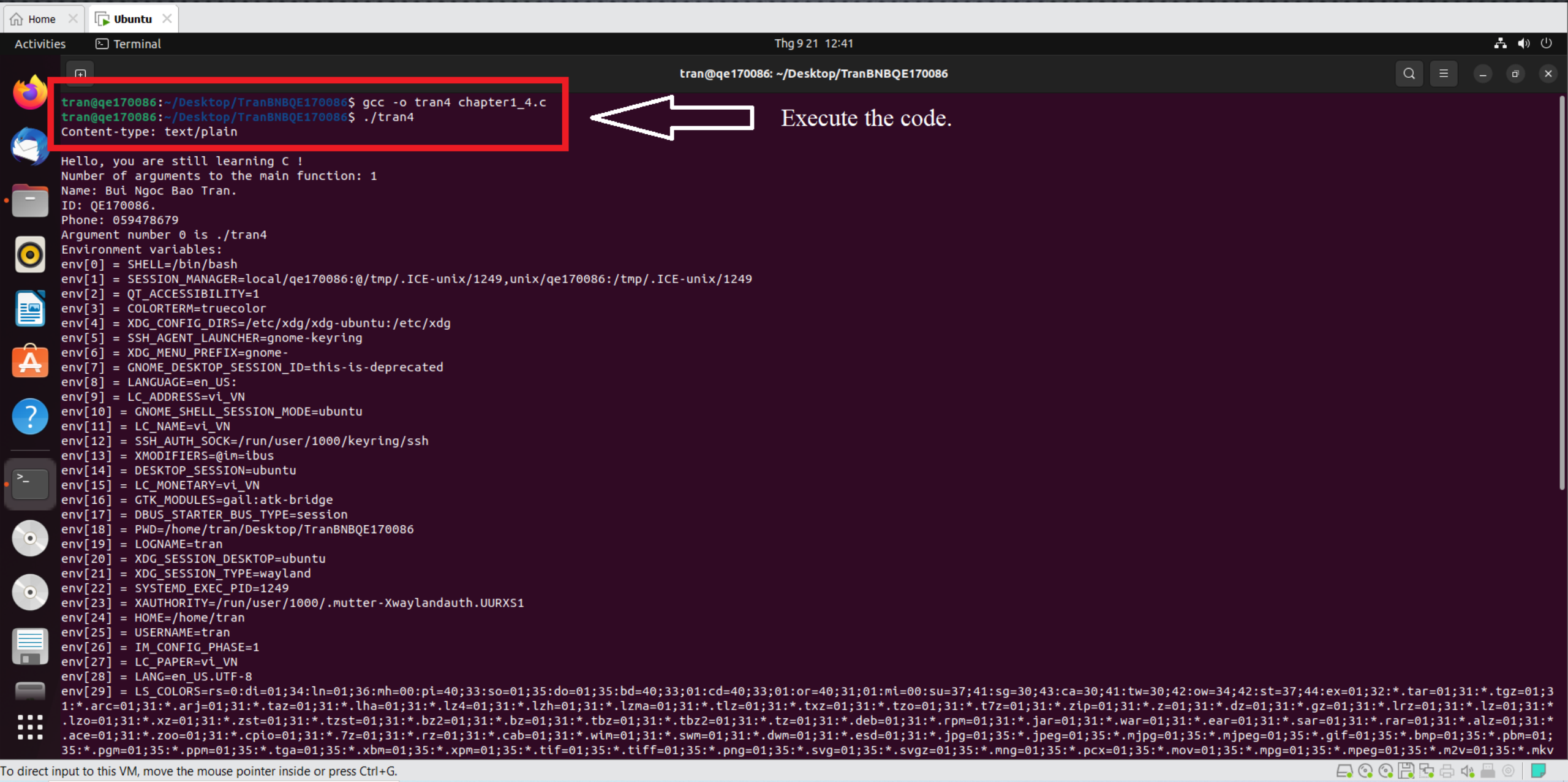
Rename your file and choose "***Save***".



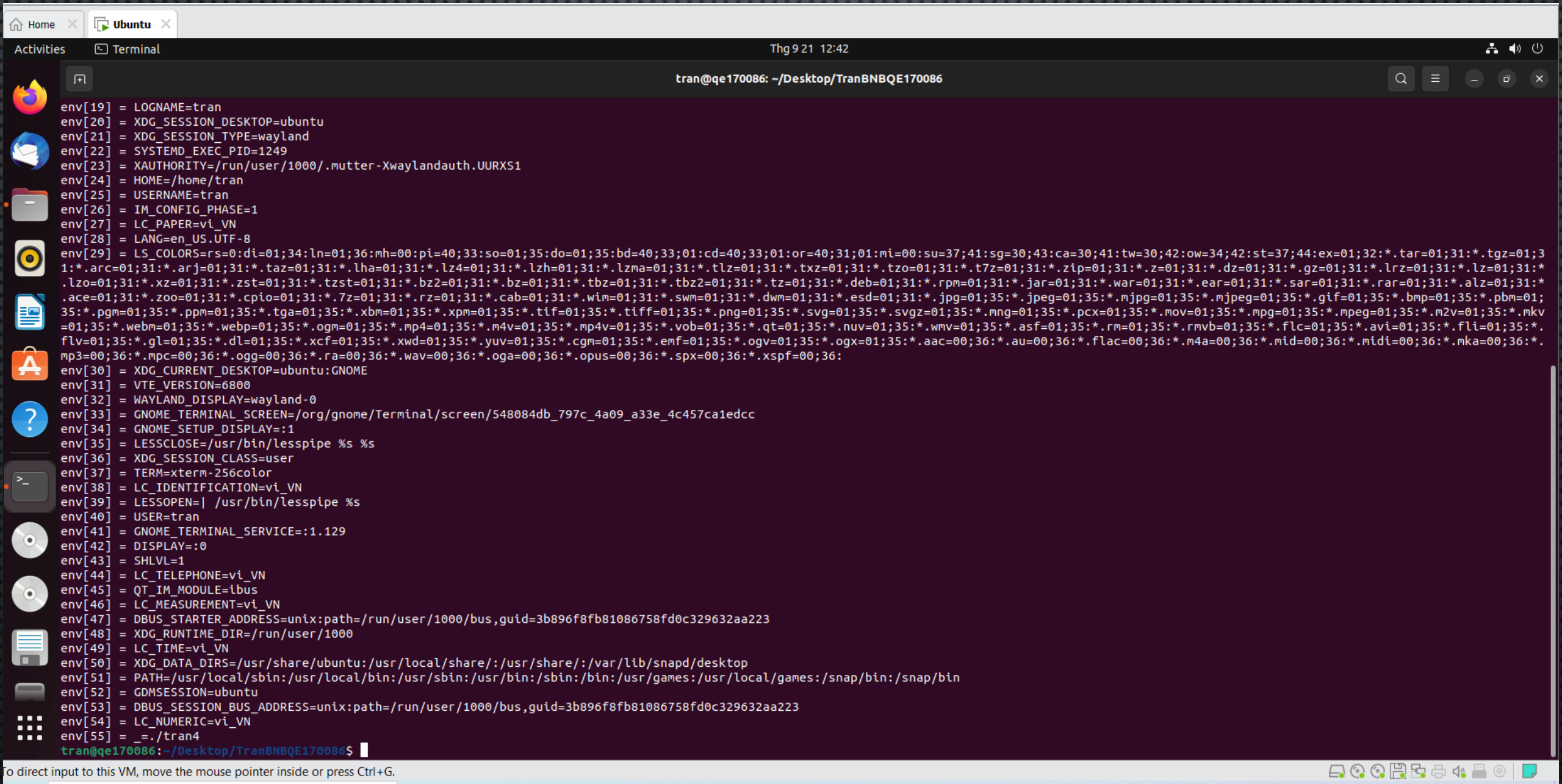
Rightclick mouse and choose "***Open in Terminal***".



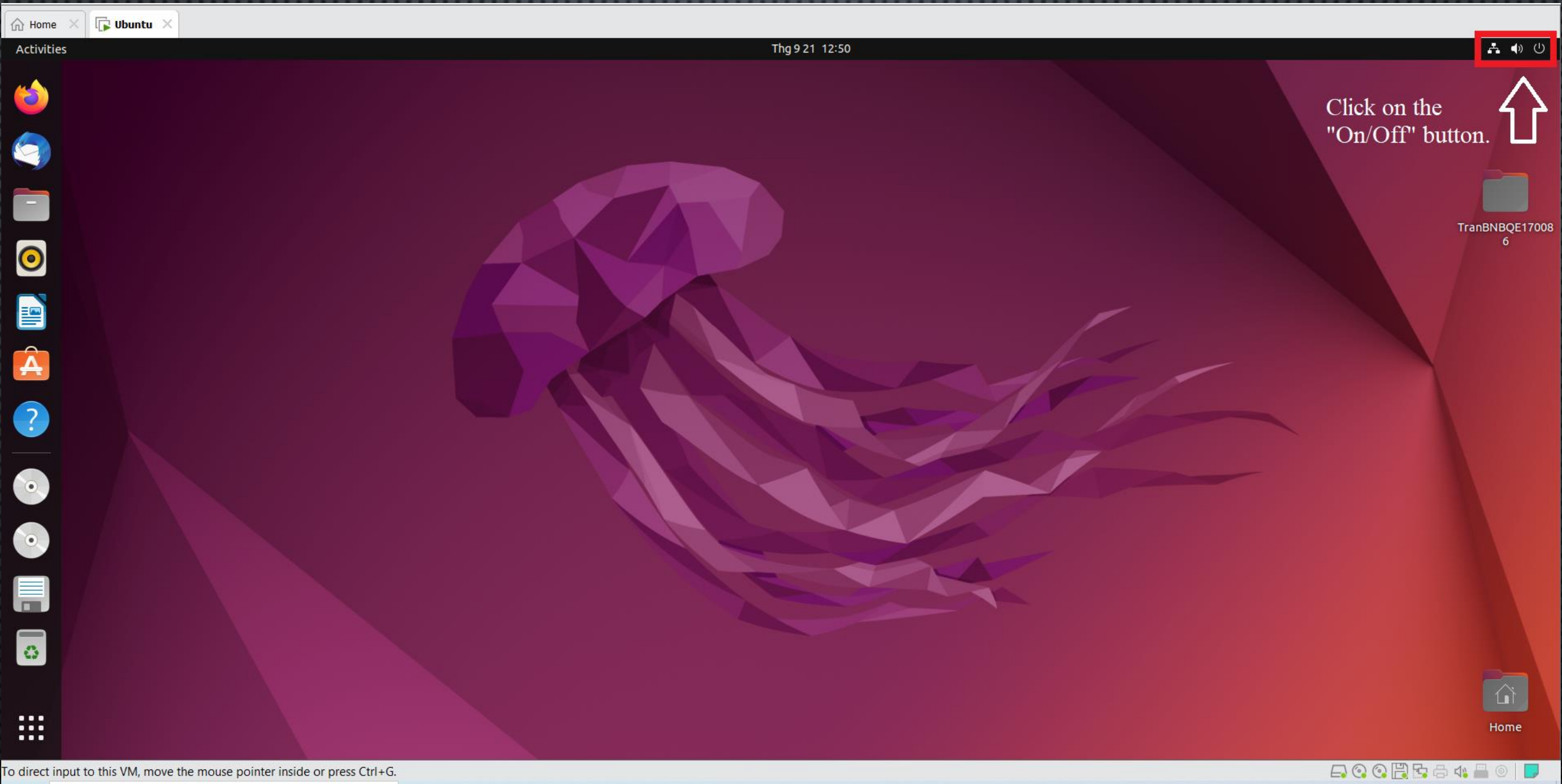
Execute the code.



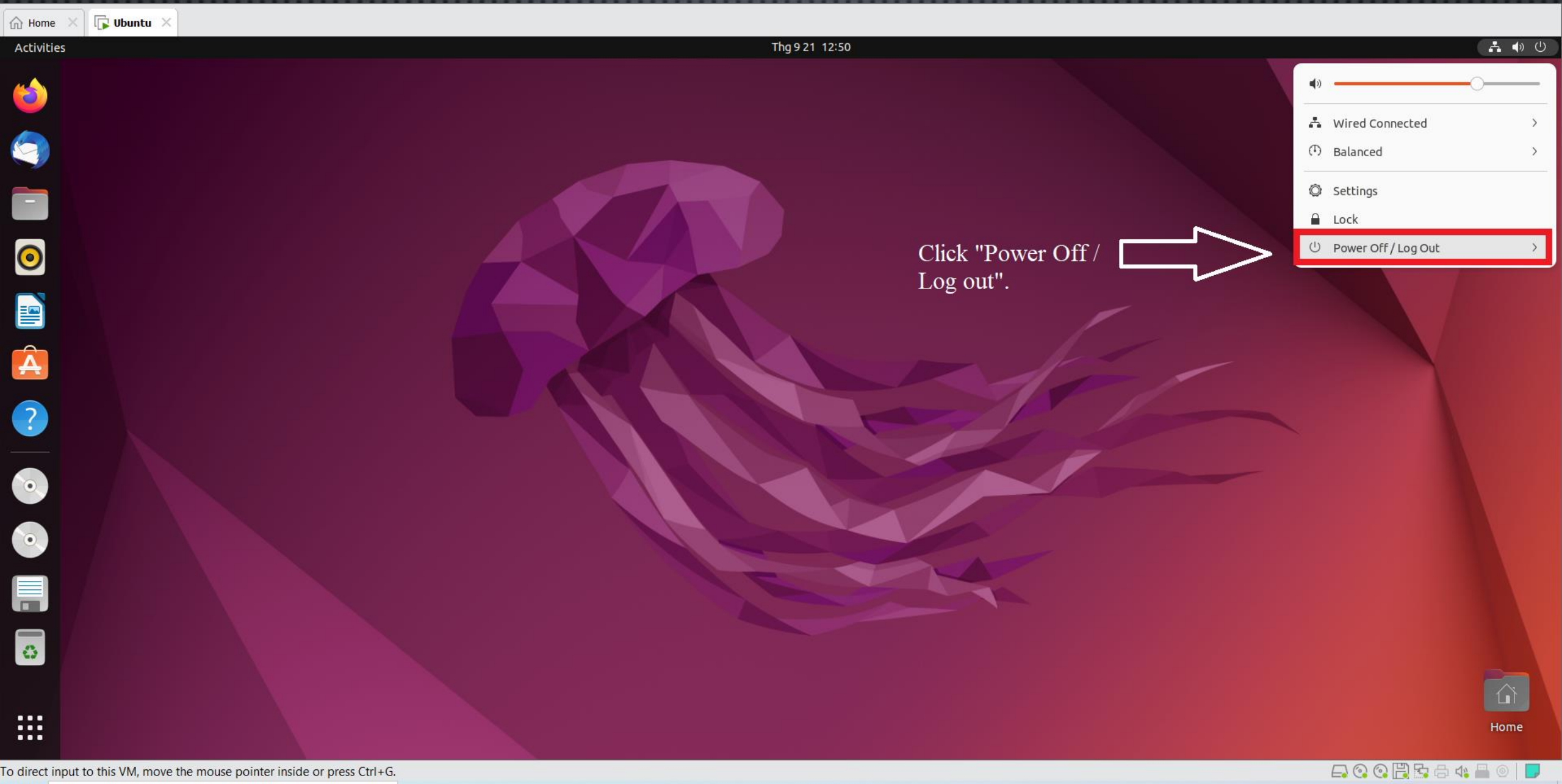
The code result.



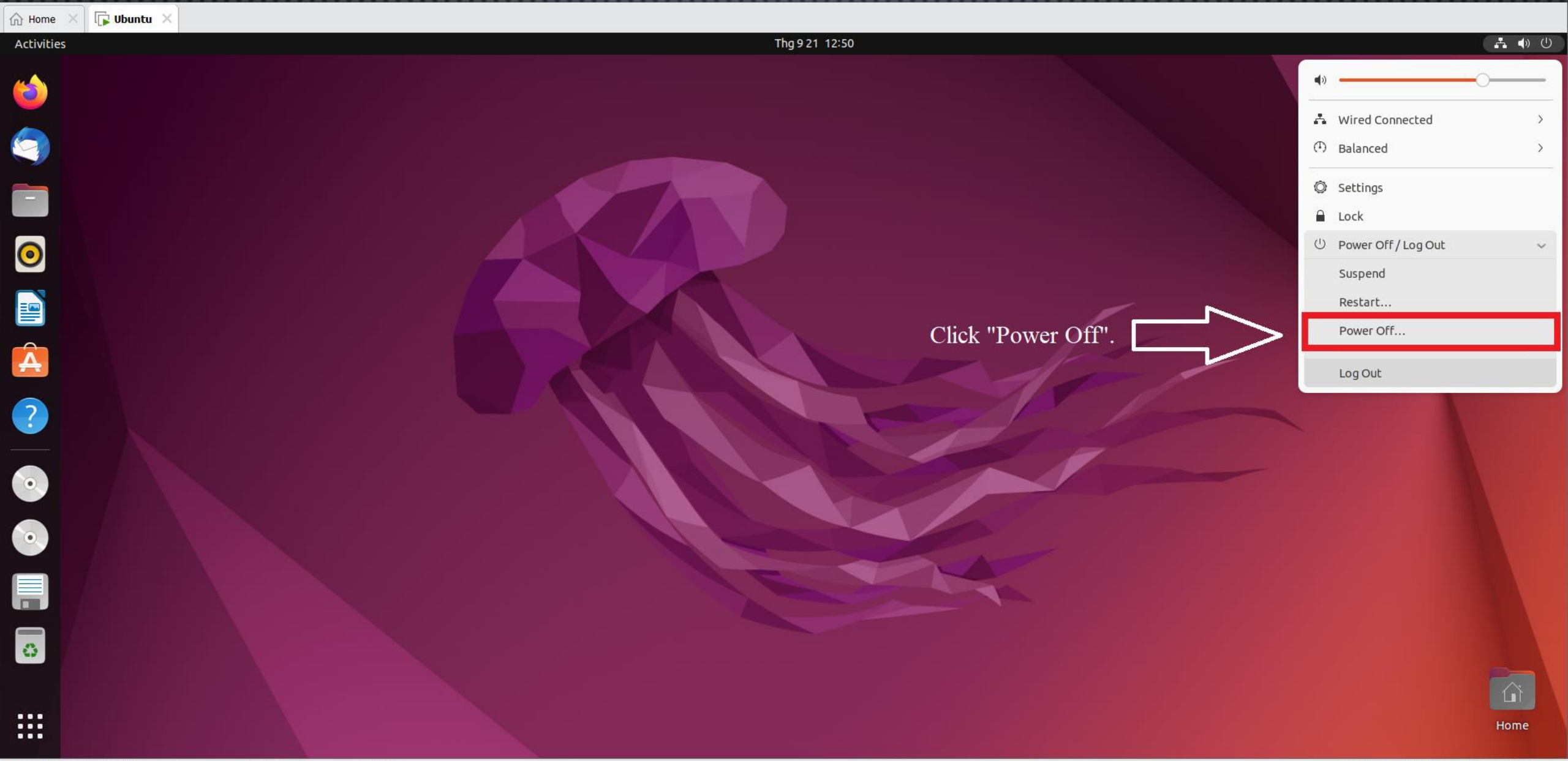
Click on the "***On/Off***" button.



Click "***Power Off / Log out***".



Click "***Power off***" and finish using virtual machine.



THE END

Bui Ngoc Bao Tran – QE170086